
Paper on

Collaborative Filtering for Implicit Feedback Datasets

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IEEE International Conference on Data Mining (2008).

(Citation 1883회, 구글 스칼라 기준)

요약 : 박종익

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1. Introduction

Recommender System Strategies

[Content based approach](#)

[Collaborative filtering approach](#)

1. Introduction

Recommender System Dataset Type

Explicit feedback dataset

Rating

Order

Implicit feedback dataset

Purchase history

Browsing history

Search Patterns

Mouse movement

1. Introduction

Implicit feedback dataset's characteristics

- No negative feedback
- Inherently noisy
- Indicates confidence
- Appropriate measures

2. Preliminaries

Observations

- r_{ui} : indexing letter (user u , v and item i , j)
- Explicit feedback dataset case
 - ~ preference
 - ~ relatively few known ratings while ignoring the missing ones.
- Implicit feedback dataset case
 - ~ user action
 - ~ natural assign values to all r_{ui} variables (no action : $r_{ui} = 0$)
- e.g) TV show recommender system
 - r_{ui} : how many times u fully watched show i
 - user watched show twice ~ $r_{ui} = 2$

3. Previous work

Collaborative filtering approach

Neighborhood models

- Item간 관계 또는 사용자들간 관계를 계산하는 방식
- Item oriented approach : “neighboring” item 점수기준
- User oriented approach : Figure 1 참고.

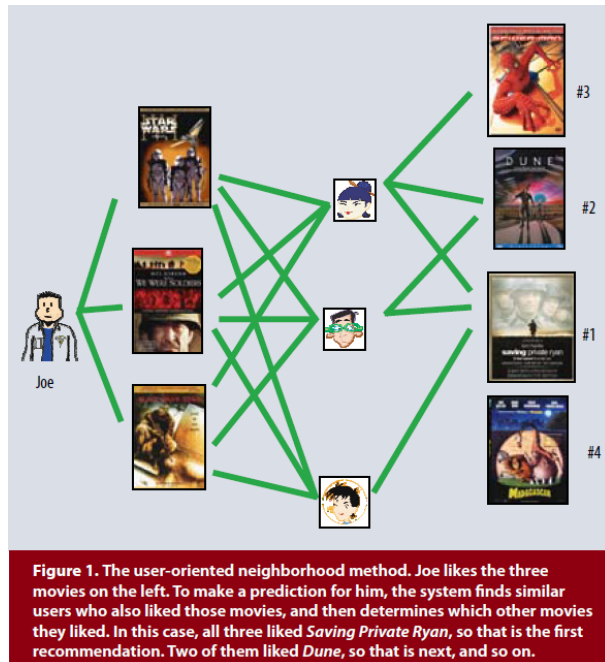


Figure1

Latent factor models

- Rating 패턴을 factor를 추출하는 방식
- 단, 해석하기 어려운 factor가 추출될 될 수도 있음
- Figure 2 참고

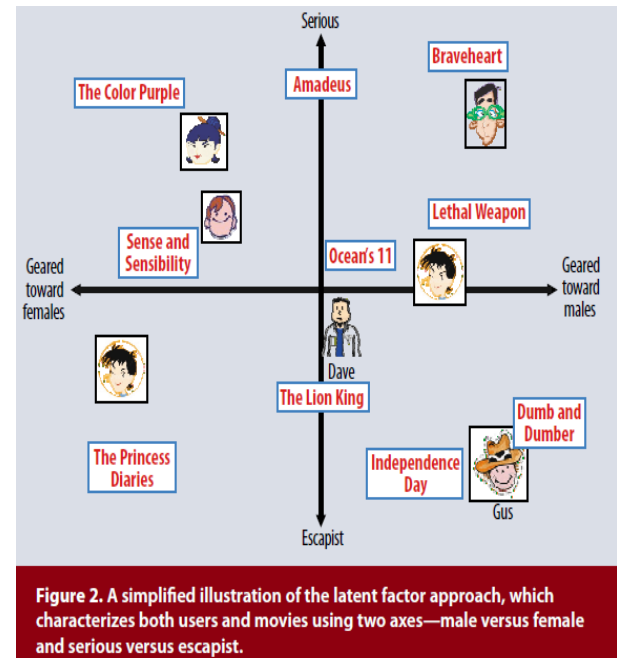


Figure2

3. Previous work

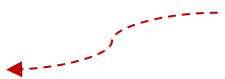
Neighborhood models

s_{ij} : similarity of i and j (fq. Pearson correlation coefficient)

$s^k(i; u)$: set of k neighbors

$$\hat{r}_{ui} = \frac{\sum_{j \in S^k(i; u)} s_{ij} r_{uj}}{\sum_{j \in S^k(i; u)} s_{ij}}$$

weighted average of the ratings for
neighboring items



3. Previous work

Latent factor models

Singular Value Decomposition (SVD)

User factor vector $\sim x_u \in R^f$

Item factor vector $\sim y_i \in R^f$

User and Item inner product $\sim \hat{r}_{ui} = x_u^T y_i$

Objective function

$$\min_{x_*, y_*} \sum_{r_{u,i} \text{ is known}} (r_{ui} - x_u^T y_i)^2 + \lambda(\|x_u\|^2 + \|y_i\|^2)$$

Regularized model

- avoid overfitting
- λ : parameter which is regularizing the model
(fq. learnt by Stochastic Gradient Descent)

4. Our model

Variable measures

$$p_{ui} = \begin{cases} 1 & r_{ui} > 0 \\ 0 & r_{ui} = 0 \end{cases}$$

Indicate Preference, binarizing the r_{ui} ,
But data zero values of $p_{ui} \rightarrow$ low level confidence

$$c_{ui} = 1 + \alpha r_{ui}$$

~ Confidence Level, also use
 $c_{ui} = 1 + \alpha \log(1 + r_{ui}/\epsilon)$

4. Our model

Objective function

Our goal : find $x_u \in R^f, y_i \in R^f$

Preference $\sim p_{ui} = x_u^T y_i$

Objective function

$$\min_{x_*, y_*} \sum_{u,i} c_{ui} (p_{ui} - x_u^T y_i)^2 + \lambda \left(\sum_u \|x_u\|^2 + \sum_i \|y_i\|^2 \right)$$

Regularized model

- avoid overfitting
- λ : data-dependent and determined by cross validation

* c_{ui} : Constant value

* Optimization Techniques : SGD, ALS, Matrix Factorization

5. Experimental study

Data description

- Summary
 - Data : digital television service (anonymization data)
 - Size : 300,000 set top boxes
 - Features : channels, timestamp
 - # of Unique TV shows = 17,000
- Training set
 - r_{ui} : # of watching minutes during a four week period
 - # of non zero $r_{ui} = 32,000,000$
- Test set
 - r_{ui}^t : channel tune events during the single week
 - $r_{ui}^t < 0.5 \rightarrow 0$ (more accurate)
 - # of non zero $r_{ui}^t = 2,000,000$

5. Experimental study

Data preprocessing

- Log scaling
 - Reason : watch the same program repeatedly
 - $c_{ui} = 1 + \alpha \log(1 + r_{ui}/\epsilon)$
 - $\epsilon = 10^{-8}$
- Down weight (adjustment)
 - Reason : watch the single channel for many hour $\rightarrow$ momentum effect
 - t-th show after a channel tune $\rightarrow \frac{e^{-(at-b)}}{1+e^{-(at-b)}}$: down weight
 - $a = 2, b = 6$ (work well and is intuitive)

5. Experimental study

Evaluation methodology

- Evaluation : each user's ordered list of the shows (most prefer = number 1)
- Methodology
 - $rank_{ui}$: the **percentile**-ranking of program i for user u

$$\overline{rank} = \frac{\sum_{u,i} r_{ui}^t rank_{ui}}{\sum_{u,i} r_{ui}^t}$$

- Lower values of $\overline{rank}$ are more desirable
- Random prediction = 50%

5. Experimental study

Evaluation result

- Our model : different # of factors, range 10 – 200
- Methodology : comparing factor model with two other models
 - Based on popularity (Baseline model)
 - Based on item – item (Neighborhood)
- Experiment
 - Rank
 - Probability (Top x% of recommended shows)
 - Performance of the factor model
 - Utility

5. Experimental study

Experiment 1 - Rank

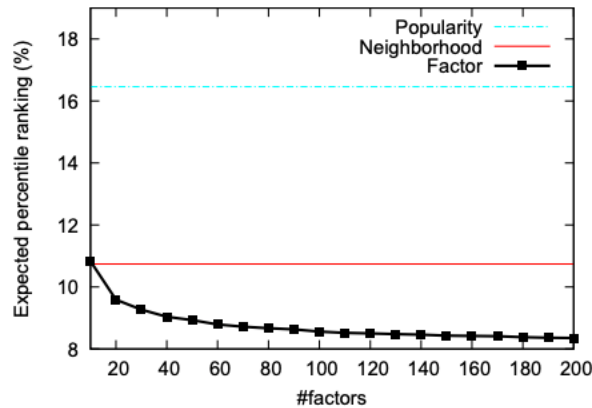


Figure 1. Comparing factor model with popularity ranking and neighborhood model.

Experiment 2 - Probability

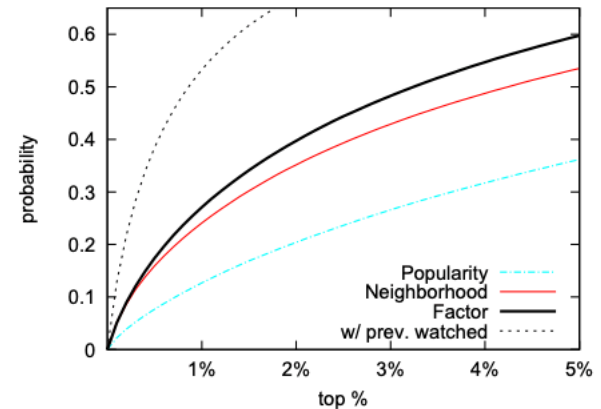


Figure 2. Cumulative distribution function of the probability that a show watched in the test set falls within top x% of recommended shows.

5. Experimental study

Experiment 3 - Performance

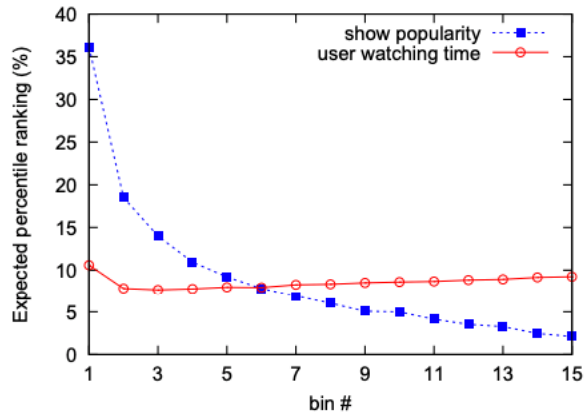


Figure 3. Analyzing the performance of the factor model by segregating users/shows based on different criteria.

Experiment 4 - Utility

So You Think You Can Dance	Spider-Man	Life In The E.R.
Hell's Kitchen	Batman: The Series	Adoption Stories
Access Hollywood	Superman: The Series	Deliver Me
Judge Judy	Pinky and The Brain	Baby Diaries
Moment of Truth	Power Rangers	I Lost It!
Don't Forget the Lyrics	The Legend of Tarzan	Bringing Home Baby
Total Rec = 36%	Total Rec = 40%	Total Rec = 35%

Table 1. Three recommendations with explanations for a single user in our study. Each recommended show is recommended due to a unique set of already-watched shows by this user.

Appendix. Learning Algorithms

■ Stochastic gradient descent

- r_{ui} 를 예측하고 예측 에러를 계산
- Gradient의 반대 방향에서 r 에 비례하는 양 만큼 파라미터를 수정

$$e_{ui} \stackrel{\text{def}}{=} r_{ui} - q_i^T p_u.$$

- $q_i \leftarrow q_i + \gamma \cdot (e_{ui} \cdot p_u - \lambda \cdot q_i)$
- $p_u \leftarrow p_u + \gamma \cdot (e_{ui} \cdot q_i - \lambda \cdot p_u)$

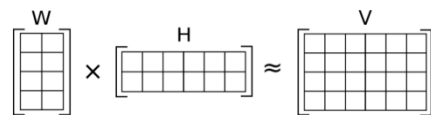
■ Alternating least squares

- q_i 와 p_u 는 unknown parameter 이기 때문에, 앞의 식(2)는 convex 가 아님
- unknown parameter 중 하나를 fix하면, 이 최적화 문제는 2차식(quadratic)이 됨
- ALS 테크닉이란 q_i 와 p_u 를 교대로 fixing하는 방식을 의미
 - ✓ p_u 를 fix하면, least-square 문제를 해결하는 방식으로 q_i 을 재 계산함
 - ✓ 보통은 SGD가 ALS보다 빠르고 용이하나, 다음 두 경우에는 ALS가 우월
 - parallelization 을 사용할 때
 - implicit data에 집중할 때

Appendix. Matrix Factorization

■ Non-negative Matrix Factorization

- 원본 행렬 V 가 있다고 했을 때, 행렬 인수 분해는 이 V 행렬을 두개의 행렬로 분리 하는 것으로, 아래 그림은 원본 행렬 V 를 $V=W*H$ 로 분리한 예이다



행렬 W는 다음과 같은 모양을 가지게 되고

책제목	특징1	특징2	특징3	특징4
협상의법칙	0.9	0	0.1	0.2
린스타트업	0	0.8	0	0
빅데이터	0.2	0.1	0.8	0.1

카테고리(책제목)과 특성과의 관계

행렬 H는 다음과 같은 모양을 가지게 된다.

	협상	스타트업	투자	비즈니스	데이터	...	...
특징1	0.92	0	0.1	0.2	0		
특징2	0	0.85	0.5	0.3	0.3		
특징3	0	0	0.3	0	0.8		
특징4	0	0	0	0	0	...	

X

원래 특성(협상, 스타트업)과 새로운 특성과의 관계

=

책제목	협상	스타트업	투자	비즈니스	데이터	...	...
협상의법칙	0.9	0	0.3	0.8	0		
린스타트업	0	0.8	0.7	0.9	0.3		
빅데이터	0	0	0.5	0	0.8		

< 그림. 책 제목과, 그 책에 나온 단어의 TFIDF 값으로 이루어진 행렬 V >

Appendix. Singular Value Decomposition

■ Low rank assumption

- 모든 matrix는 두 matrix의 곱으로 표현이 가능함
- 이때 matrix의 rank 가 작다면 두 matrix의 rank 역시 더 작은 형태로 표현이 가능함
- n by m matrix R 이 n 이나 m 보다 작은 k 만큼의 rank를 가졌을 때, R 은 n by k matrix P 와 m by k matrix Q 의 곱으로 표현할 수 있음. 즉 $R=PQ^T$ 으로 표현됨 $r_{ui} = p_u \cdot q_i$
- User u 가 item i 의 점수를 주는 방식은 user u 의 item들에 대한 숨겨진 interest p_u 와 그에 대응하는 item들의 숨겨진 특성 q_i 에 의해 결정된다는 사실을 알 수 있음

